

Hemorrhoids

Clinical Presentation

- Rectal bleeding (bright blood at the outer part of stool after defecation), burning, pain, itch, and swelling (which causes pruritus) with possible prolapse of anal tissue.
- It is due to distal displacement of the hemorrhoidal cushions, with venous distension, protrusion, bleeding, and possible thrombosis.

Classification of Hemorrhoids

- **External:** painful and develop below the dentate line.
- **Internal:** develops above the dentate line, may cause bleeding, and usually painless unless complicated. Internal hemorrhoids can be classified into:

Grade I	asymptomatic outgrowth of anal mucosa due to straining.
Grade II	protrusion of a small piece of the anal mucosa during bowel movement with spontaneous return to position after defecation.
Grade III	the protrusion remains after defecation, but it can be returned manually. Bleeding is common.
Grade IV	permanent bulge that cannot be returned even manually, very painful, and at high risk of complications.

- Mixed

Risk Factors

- **Bowel habits:** Factors affecting the frequency, consistency, and ease of evacuation such as constipation, and IBS (causes frequent, and prolonged toileting), increase the risk of hemorrhoid.
- Age: peak between 45 and 65.
- Lifestyle: work and activities associated with exertion and straining.
- Diet: fiber-depleted diet that is high in sugar and white flour.
- Other factors: pregnancy (especially late in pregnancy and improves postpartum), and spinal cord injury.

Red Flags

- Acute onset pain (may suggest thrombosed external hemorrhoids or other comorbidities).
- Severe pain when defecating.
- Bleeding: black tarry stool or fresh blood in the stool.
- Pain and protrusion (Grade III or IV).
- If there is no relief of symptoms within 7 days after starting therapy, if symptoms deteriorate, or if the patient develops severe pain/bleeding.

Prevention of Hemorrhoids

Avoid constipation (the most important measure), and avoid straining and limit toileting time to 1-2 minutes per bowel movement.

Management of Hemorrhoids

Goals of Therapy

Relieve symptoms, promote anal hygiene, enhance bowel habits, and prevent complications.

Non-Pharmacologic Therapy

- Fiber supplements: first line in management of hemorrhoids and is usually adequate to control grade I and II – see constipation for more information about doses, etc.
- Sitz bath: sitting in a tub of warm water for 15 minutes 3-4x/day. It cleanses the area, provides comfort, relieves pruritus and irritation, and improves healing by improving blood circulation.
- Wash the anorectal area with mild soap and water and wipe, rinse, and pat dry gently or wipe with wet tissue or soft cloth.
- Replace prolapsed tissue using wet cloth.
- Rubber-banding.
- Injection sclerotherapy: reduces bleeding especially in patients with secondary hemorrhoids and patients with reduced immunity.
- Electrocoagulation therapies (e.g., infrared ablation, and bipolar diathermy).
- Surgical hemorrhoidectomy: reserved for patients with grade III and IV hemorrhoids particularly those with recurrent, refractory, or thrombosed hemorrhoids.

Pharmacologic Therapy

- Only symptomatic treatments are available. The reduce pain, burning, itch, swelling, discomfort, and irritation, but with no effect on bleeding or prolapse.
- Used every morning and evening and after each bowel movement during the day as needed.
- Ointments and creams are preferred over suppositories and easier to apply but consider the patient's preference.
- Non-prescription hemorrhoids drugs are used for patients ≥ 12 years of age.

Modality	Notes
Topical anaesthetics Agents: dibucaine (cinchocaine) and pramoxine.	▪ Relieve pain but should be used for < 7 days to avoid dermatitis (pramoxine is associated with less dermatitis compared to other agents). ▪ Faster onset of action but shorter duration of effect compared to other agents. ▪ Available in combination with other symptomatic treatments ▪ Adverse reactions: local burning and itching. Minimal systemic adverse reactions unless used excessively.
Topical Anti-inflammatory Agents: HC	▪ Relieves anal itch and inflammation. Limit the treatment to 7-14 days. ▪ The onset of action is 12 hours but lasts longer than local anesthetics. ▪ Adverse reactions: skin atrophy with prolonged use and may mask infections.
Astringents Agents: hamamelis, and zinc sulfate.	▪ Promote protein aggregation (clumping) and skin dryness and thus relieve burning, itching, swelling, and pain but only effective in mild cases (hydrocortisone is required if there is significant itch, or discharge). ▪ Adverse reactions are insignificant.
Protectants Agents: glycerin, and white petrolatum.	▪ Lubricate (ease the passage of stool) and form a barrier that prevents the contact of perianal tissue with the stool and thus reduce irritation, itch, pain and burning. ▪ Adverse reactions are insignificant.
Vasoconstrictors Agents: phenylephrine.	▪ Reduce bleeding. ▪ The onset of action is a few seconds to 1 min with a duration of 2–3 hours. ▪ May have the same drug interactions of other sympathomimetics. ▪ Adverse reactions are insignificant but may cause systemic adverse reactions if applied to broken skin.
Phlebotonics (venotonics) Agents: hesperidin, citrus bioflavonoids, and diosmin.	▪ These are oral agents. ▪ Strengthen the vein walls in anorectal tissue, so they improve bleeding, reduce inflammation, pruritis, and discharge in patients with grade I and II. ▪ Diosmin may increase the effect of anticoagulants (use with caution). ▪ Adverse reactions: nausea, abdominal pain, diarrhea, and itchiness

Other Modalities

- Acetaminophen can be used to control pain (avoid opioids because of constipation).
- Framycetin is used to reduce/prevent secondary bacterial infections, swelling, and inflammation.
- Bovine cartilage 5% cream is used to relieve itch.
- Yeast has insufficient evidence.
- Horse chestnut reduces swelling and inflammation. Avoid during pregnancy, breastfeeding, and in patients with bleeding disorders.

Pregnancy

- Hemorrhoids may get worse during pregnancy.
- Avoiding constipation (to prevent hemorrhoids) and increasing fiber intake (to treat hemorrhoids) are the first choices. Laxatives, increasing fluid intake, and toilet training can be used to correct constipation. Sitz bath can also be used.
- No safety data is available for symptomatic treatments, but they can be used, preferably external agents, for short term to promote comfort during the healing of hemorrhoids. Local anesthetics and hydrocortisone can be used under medical supervision.
- Ointments and creams are preferred for external hemorrhoids while suppositories are preferred for internal hemorrhoids.

Patients Education

- When using suppositories: Do not insert suppositories too far into the rectum and lie down for few minutes after inserting the suppository to avoid slipping into the rectum.
- When using ointments and creams:
 - If used externally: apply sparingly and do not insert into the rectum.
 - If used internally: lubricate the applicator with the cream or the ointment before insertion.

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